INDIAN INSTITUTE OF INFORMATION TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
COURSE: IMAGE AND VIDEO PROCESSING (IVP)

DeepGuard: Test Plan & Test Data Documentation

Proposed Verification Strategy, Benchmark Datasets & Expected Evaluation Criteria
[Initial Project Planning Submission]

Project Group: IVP_Group-10_DeepGuard

Team Members:
IIT2024108 | IIT2024109 | IIT2024110
IIT2024115 | IIT2024119 | IIT2024122

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Document Control & Test Planning Metadata

Document Version	Project Phase	Primary Authors	Scope Summary
1.0 (Current)	Project Proposal & Planning	IVP Group 10 Team	Initial test plan, proposed evaluation datasets, anti-leakage partitioning, and planned test cases.

1. Testing Objectives and Verification Scope

The objective of this Test Plan is to define the testing methodologies, dataset preparation protocols, and evaluation criteria that will verify the DeepGuard deepfake video detection system once implemented. The proposed testing framework will evaluate: (1) video-level and frame-level detection accuracy across authentic and synthetic media; (2) cross-manipulation transfer across four distinct forgery families (Deepfakes, Face2Face, FaceSwap, NeuralTextures); (3) zero-shot cross-dataset generalization on Celeb-DF v2; (4) robustness under social media perturbations (compression, noise, blurring, downsampling); and (5) operational latency and GPU VRAM footprint on commodity hardware.

2. Proposed Benchmark Datasets & Storage Strategy

The evaluation strategy will rely on three standardized deepfake video benchmarks:

Dataset Name	Compression Tier	Video Volume	Proposed Evaluation Role	Official Access / Provenance
FaceForensics++ (FF++)	C23 (HQ, H.264)	7,000 videos (16.69 GB) 1,000 real + 4,000 fake	Primary in-domain training, validation & 4-fold Leave-One-Family-Out matrix	TUM Computer Vision Group (Academic request required)
Celeb-DF v2	Standard YouTube	6,529 videos total 518 official test videos	Independent zero-shot cross-dataset generalization benchmark	Celeb-DF Project (Research access required)
DFDC (Challenge)	Kaggle / Meta	~100,000 videos (~470 GB)	Independent cross-dataset benchmark (Subject to Kaggle credentials)	Kaggle Deepfake Detection Challenge

3. Data Preparation & Anti-Leakage Partitioning Protocol

To guarantee scientific integrity and prevent data leakage, the project will enforce strict subject-level partitioning: In deepfake video datasets, multiple manipulated videos are synthesized from the same underlying source video. Splitting videos randomly allows the same actor to appear in both training and test sets, artificially inflating detection scores. DeepGuard will assign an indivisible group_id to every pristine video and all of its manipulated counterparts. All partitions (train, validation, test) will be formed by splitting disjoint sets of group_ids. For FaceForensics++ C23, the 1,000 subjects will be split into 720 training subjects (5,040 videos), 140 validation subjects (980 videos), and 140 test subjects (980 videos).

4. Planned Functional & Stress Test Cases

The testing framework will implement the following structured test cases during the verification phase:

Test ID	Test Title & Objective	Input Specification	Expected Success Target
TC-001	Authentic Video Specificity Test: Verify detection of clean real videos without false alarms.	Prisitne videos from FF++ C23 and Celeb-DF v2 test splits.	Clean specificity >= 90.0% (False Positive Rate <= 10.0%) under high-sensitivity threshold.
TC-002	Whole-Face Replacement Test: Verify detection of deepfakes with boundary seams (Deepfakes, FaceSwap).	Manipulated videos generated by autoencoder swaps and graphics blending.	Detection accuracy >= 95.0% and ROC-AUC >= 0.95.
TC-003	Facial Reenactment Test: Verify detection of expression reenactments (Face2Face).	Manipulated videos where only facial expressions are transferred via 3DMM.	ROC-AUC >= 0.78 on unseen Face2Face holdout fold.
TC-004	Neural Texture Synthesis Test: Verify detection of GAN-rendered mouth and teeth textures.	NeuralTextures manipulated videos from FF++ C23.	ROC-AUC >= 0.90 on unseen NeuralTextures holdout fold.
TC-005	Compression Stress Test: Verify model robustness under lossy compression.	Videos re-encoded with H.264 C40 and JPEG quality factors Q=50, 25, 10.	AUC retention >= 0.60 under severe compression; no unhandled crashes.
TC-006	Resolution Downsampling Test: Verify pipeline handling of lower resolution media.	Videos downsampled from 1080p to 720p, 480p, and 240p.	Preprocessing pipeline resizes properly to 112x112 without aspect ratio distortion.
TC-007	Occlusion & No-Face Fallback Test: Verify pipeline behavior when faces are missing or occluded.	Video clips with extreme side profiles, hand occlusions, or no detected face.	System executes fallback center cropping without raising fatal runtime exceptions.
TC-008	Bitwise Determinism Test: Verify repeated	Identical video evaluated 10	Delta logits == 0.000000 across

	inference reproducibility under fixed seed.	consecutive times under fixed seed.	all repeated forward passes.
TC-009	Model Export Parity Test: Verify numerical parity of exported ONNX runtime models.	Exported ONNX model evaluated against PyTorch checkpoint.	Maximum absolute numerical difference between outputs < 1e-5.
TC-010	Hardware VRAM & Latency Benchmark: Verify execution fits 6GB RTX 3050 constraints.	100 forward passes of 16-frame sequences on NVIDIA RTX 3050.	P50 forward latency <= 35 ms; peak VRAM allocation <= 1.0 GB.

5. Planned Evaluation Metrics & Target Success Criteria

The project will evaluate models across standardized quantitative metrics, establishing clear target criteria:

Evaluation Metric	Mathematical Formulation	Proposed Target Criterion	Research Rationale
ROC-AUC	Area Under the Receiver Operating Characteristic Curve	>= 0.90 (In-Domain FF++) >= 0.75 (Cross-Dataset)	Primary threshold-independent ranking metric for binary classification.
Classification Accuracy	(TP + TN) / Total Samples	>= 90.0% (FF++ C23)	Standard overall correctness indicator across balanced test partitions.
Clean Specificity	TN / (TN + FP)	>= 90.0% (Authentic Media)	Guarantees authentic user media is not erroneously flagged as deepfake.
Equal Error Rate (EER)	Operating point where FPR == FNR	<= 10.0% (In-Domain)	Standard biometrics benchmark assessing false accept vs false reject balance.
Expected Calibration Error	Weighted difference: sum(|conf(B) - acc(B)|)	<= 10.0% (Post-Calibration)	Ensures model confidence reflects empirical probability of correctness.
Forward Pass Latency	P50 execution time for 16 frames	<= 35.0 ms (RTX 3050 GPU)	Guarantees practical usability for near-real-time video analysis.

6. Illustrative Evaluation Output Template

Note: The following table represents an illustrative evaluation template that will be populated during the testing phase. The values shown below are illustrative placeholders, NOT actual project measurements.

Video ID / Filename	Ground Truth	Predicted Label	Calibrated Prob (p_cal)	OOD Flag	Execution Status
sample_real_001.mp4	REAL	REAL (Illustrative)	0.084 (Illustrative)	False	PLANNED FOR TESTING
sample_real_002.mp4	REAL	REAL (Illustrative)	0.112 (Illustrative)	False	PLANNED FOR TESTING
sample_fake_df_001.mp4	FAKE (Deepfakes)	FAKE (Illustrative)	0.942 (Illustrative)	False	PLANNED FOR TESTING
sample_fake_f2f_001.mp4	FAKE (Face2Face)	FAKE (Illustrative)	0.825 (Illustrative)	False	PLANNED FOR TESTING
sample_fake_fs_001.mp4	FAKE (FaceSwap)	FAKE (Illustrative)	0.961 (Illustrative)	False	PLANNED FOR TESTING
sample_fake_nt_001.mp4	FAKE (NeuralTextures)	FAKE (Illustrative)	0.893 (Illustrative)	False	PLANNED FOR TESTING